

NLP-Powered Agentic AI System for News Analysis

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1. Introduction

Recent advances in Artificial Intelligence have transformed traditional NLP systems into Agentic AI systems capable of autonomous reasoning, planning, memory management, and dynamic tool selection. Unlike conventional NLP models that perform a single predefined task, agentic systems can analyze user goals, decide which tools are required, and execute multiple actions automatically.

The objective of this project is to develop an NLP-powered Agentic AI system that combines News Classification and Named Entity Recognition (NER) within an autonomous agent framework. The proposed system demonstrates perception, planning, memory, and action execution to fulfill user requests intelligently.

2. Dataset Selection and Justification

The AG News dataset was selected for this project. The dataset contains approximately 120,000 news articles categorized into four classes:

- Business
- Sports

- World
- Science/Technology

The dataset was chosen because it is large, balanced, and suitable for multiple NLP tasks. Each record contains a news title and description, making it appropriate for text classification and information extraction tasks.

Dataset Statistics:

- Total Records: 120,000
- Categories: 4
- Records per Category: Approximately 30,000

The balanced distribution reduces class imbalance issues and improves evaluation reliability.

3. Data Preprocessing

Several preprocessing steps were performed before implementing NLP tasks.

3.1 Dataset Loading

The AG News dataset was loaded using Python and Pandas.

3.2 Label Mapping

Numeric category labels were converted into readable class names:

- Business
- Sports
- World
- Sci/Tech

3.3 Text Construction

The news title and description were merged into a single text field to provide richer contextual information.

3.4 Dataset Sampling

A subset of 2,000 records was selected for experimentation and testing purposes.

3.5 Train-Test Split

The dataset was divided into:

- Training Set: 1600 samples
 - Testing Set: 400 samples
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4. NLP Tasks

4.1 Task 1 – News Classification

A news classification component was implemented to categorize news articles into one of four categories:

- Business
- Sports
- World
- Science/Technology

The classifier analyzes textual content and predicts the most relevant category.

Example:

Input:

"Apple released a new AI software platform."

Output:

"Sci/Tech"

The classification module serves as one of the core NLP tools used by the agent.

4.2 Task 2 – Named Entity Recognition (NER)

Named Entity Recognition was implemented using a transformer-based BERT model.

The NER model extracts important entities from textual input, including:

- Persons
- Organizations
- Locations

Example:

Input:

"Apple CEO Tim Cook visited London."

Output:

Apple → Organization

Tim Cook → Person

London → Location

Microsoft → Organization



Figure 1 demonstrates the successful extraction of named entities using the transformer-based NER model.

5. Agentic AI Architecture

The proposed system follows an Agentic AI architecture consisting of the following components:

Perception Layer

Receives and interprets user input.

Planner

Determines which NLP tools are required.

Tool Selection Module

Selects Classification, NER, or both tools dynamically.

Memory Component

Stores previous interactions and intermediate results.

Action Module

Executes NLP tools.

Response Generator

Combines outputs and generates the final response.

ARCHITECTURE



Figure 2 illustrates the overall architecture of the proposed Agentic AI system.

6. Decision-Making Logic

The system employs a rule-based planning mechanism.

The planner analyzes user intent and selects the required tools:

- Entity-related request → NER
- Category-related request → Classification
- General analysis request → Both tools

This enables autonomous task decomposition and dynamic tool selection.

7. Example Interaction 1

User Query:

"Apple announced a new AI software in London."

Agent Decision:

Plan: BOTH

Classification Result:

Sci/Tech

NER Result:

- Apple → Organization
- London → Location



Figure 3 shows the agent automatically selecting multiple NLP tools and generating the final response.

8. Example Interaction 2

User Query:

"Pakistan won the cricket match yesterday."

Agent Decision:

Plan: CLASSIFICATION

Classification Result:

Sports



```
{4}
✓ 0s
if plan == "CLASSIFICATION":
    result["plan"] = "CLASSIFICATION"
    result["category"] = classify_news(user_input)
else:
    result["plan"] = plan

return result

print(agent("Pakistan won the cricket match yesterday"))

>>> {'plan': 'BOTH'}
```

Figure 4 demonstrates dynamic tool selection for a sports-related query.

9. Results and Analysis

The developed system successfully demonstrated:

- Autonomous planning
- Dynamic tool selection
- News classification
- Named Entity Recognition
- Memory-based interaction handling

The system was able to analyze user goals and select appropriate NLP tools without manual intervention.

10. Challenges Encountered

Several challenges were encountered during implementation:

- Transformer version compatibility issues

- HuggingFace dataset loading problems
- Summarization pipeline compatibility limitations
- Computational resource constraints

These issues were resolved through environment adjustments and alternative NLP approaches.

11. Ethical Considerations

Potential ethical concerns include:

- Incorrect classifications
- Entity recognition errors
- Model bias
- Hallucinated outputs
- Over-reliance on automated decision-making

Human oversight remains important when deploying agentic systems in real-world environments.

12. Conclusion

This project successfully implemented an NLP-powered Agentic AI system capable of perception, planning, memory management, and dynamic tool selection. The integration of News Classification and Named Entity Recognition demonstrated how multiple NLP tools can work together within an agentic framework. The project highlights the advantages of Agentic AI over traditional NLP systems and provides a foundation for future improvements through advanced transformer models and LLM-based planning mechanisms.